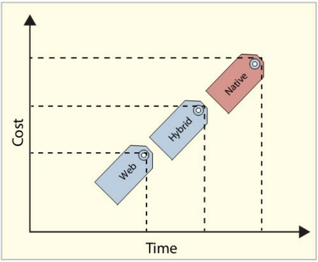

Figure 3: Native vs Hybrid vs Web

features. In case of cross platform development, it is recommended to harness the common business logic layer among platform-specific builds.

- *Android native layer:* This layer takes care of platform-specific capabilities (camera, database, HTTP communication, persistent store, etc) to be used by Android mobile applications. This layer is not accessible with Web application implementation. This is the reason Web applications possess limited access to mobile platform capabilities.

Mobile device management

An enterprise application must provide data security and real-time management of the Android application/device. Mobile device management (MDM) is the approach for user device management of data security and real-time device management. The following are some of the imperative features offered by MDM:

- Device access management
- Managing installed applications
- Enabling/disabling device features/flows
- Identifying and managing security controls
- Real-time device monitoring/tracking locations
- Remote locking/wiping of devices
- Policy management

MDM solutions or products possess both server-side and Android mobile client-side handling. Server consoles have features to manage and control devices with their respective configurations. These configurations or settings are communicated to Android client devices via their respective protocols. And the relevant client implementation takes care of configuration management and other device level security mechanisms.

Mobile application management

Mobile application management (MAM) caters to an enterprise's data security and application management needs. As the term indicates, MAM emphasises on application level management, rather than management at the device level. The emergence of BYOD/CYOD has led to the development of MAM, because users prefer to not carry enterprise MDM systems on their personal devices. However, enterprise application-specific security mechanisms and policies can be accepted by users. The following are some of the imperative features offered by MAM:

- Application access management
- Managing the application's life cycle
- Enabling/disabling application features/business flows
- Ensuring application data security while data is at rest
- Real-time application monitoring
- Data wiping while on the go
- Remote resuming/suspending of applications

This solutions approach requires a server-side system and the respective Android client-side handling to achieve the desired features. A server-side system, along with the admin console and the client at the device, communicates via ubiquitous protocols. These protocols provide controls at the server for managing enterprise applications.

Comparing native, Web and hybrid apps

The amalgamation of best-of-breed technologies is known as a mobile hybrid application. Figure 3 illustrates hybrid vs native vs Web applications on a scale of time and cost.

Availability of diverse mobile platforms is one of the challenges for any enterprise application. Cross-platform support is a must for any mobile application today. Along with this, the application is also expected to cover platform-specific diverse form factors. This diversity in mobile devices and technology is what has resulted in mobile hybrid applications becoming a preferred implementation approach to application development. END

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By: Raghvendra Singh Dikhit

The author is a solutions architect by profession and author of the book 'Enterprise Mobility Breakthrough'. He has hands-on experience with enterprise mobility solutions, services and products. He has worked with Fortune 500 firms for mobile solutions (architecture, design, development, quality assurance and porting). He holds a master's degree in computer applications, an MBA and a master's degree in English.